Grouped Multi-Attention Network for Hyperspectral Image Spectral-Spatial Classification

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***Abstract*— Deep learning (DL) has been a powerful tool for hyperspectral image (HSI) classification. However, it is still an open issue to effectively learn highly discriminative features from HSI, due to the high-dimensionality and complex spectral-spatial characteristics. To settle this issue, we propose a new band-grouping guided multi-attention module for the performance promotion of spectral-spatial feature learning. First, based on the fact of high relevance between adjacent spectral bands and low dependencies across long-range ones, all the spectral bands are adaptively divided into multiple nonoverlapping groups where relevant bands are included. The advantage is to reduce the spectral dimension and data complexity when processing and analyzing each group. Then, a multi-attention mechanism, which not only explores the intragroup salient information but also propagates the intergroup difference information, is embedded into the convolutional neural networks (CNNs) to learn group-specific spectral-spatial features. By emphasizing useful spectral/spatial information and squeezing useless information with attention mechanism, the severability of learned features is enhanced. Based on this module, a spectral-spatial classification network is built, named by grouped multi-attention network (GMA-Net). The GMA-Net contains a two-branch architecture, i.e., pixelwise spectral feature learning and patchwise spectral-spatial feature learning. Via fusing the features from two branches, the complementary and discriminative features provided by pixelwise and patchwise learning manner can be integrated to further boost the classification performance. Experimental results demonstrate that the proposed method is superior than several state-of-the-art approaches. Codes are available at: https://github.com/luting-hnu.**

***Index Terms*— Band-grouping, deep learning (DL), hyperspectral images (HSIs) classification, multi-attention, spectral-spatial feature.**

I. Introduction

HYPERSPECTRAL image (HSI) usually contains hundreds of contiguous spectral bands, which record spectral reflectance and spatial contexture of scenes. The abundant spectral and spatial information significantly strengthen the ability of identifying different materials. As a consequence, HSI classification (HSIC), which aims to assign a unique label to each pixel, has become a hot research topic in the field of remote sensing [1]. Especially, HSIC has been applied to various applications, such as urban development [2], [3], environmental protection [4], crop monitoring [5], and national defense [6]. In the past decades, many HSIC works are developed.

In the early stage, various machine learning algorithms have been used for HSIC, including support vector machines (SVMs) [7], random forests (RF) [8], multinomial logistic regression (MLR) [9], [10], and so on. Meanwhile, carefully handcrafted spectral-spatial features are designed for better classification performance, e.g., local binary patterns (LBPs) [11], Gabor features [12] and extended morphological profiles (EMPs) [13]. More recently, some scholars have developed several new feature extractors. For instance, Hong et al. [14] developed a spatial–frequency joint feature extractor named invariant attribute profiles (IAPs), which effectively improves the robustness of modeling HSI spatial information. These methods demonstrate the effectiveness of spectral-spatial classification, but also suffer from some issues as follows. On one hand, handcrafted features are usually low-level features whose representation ability for complex spectral-spatial structures is limited. On the other hand, the design of handcrafted features depends on limited prior information, which leads to poor generalization performance for unknown datasets.

Nowadays, deep learning (DL) has achieved great success in the field of computer vision, due to its powerful ability to automatically extract high-level features from a large amount of data [15], [16], [17], [18]. Therefore, DL-based classification methods have also become a hot research direction for HSIC. With an automatic encoder, DL is applied to HSIC for the first time by Chen et al. [19]. After that, various DL models are successively applied to HSIC, including stacked autoencoders [20], [21], [22], deep belief networks [23], [24], recurrent neural networks/long short-term memory [25], [26], [27], graph convolutional networks [28], [29], [30], [31], generative adversarial networks [32], [33], [34], [35], [36], and so on. Convolutional neural network (CNN), as one of the most popular DL models for image processing and analysis, has drawn a great deal of attention in HSIC. Generally, CNN-based HSI classification methods crop a HSI patch from the original HSI and input it into the network to generate class label for the central pixel of this patch. Standard CNN

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with cascading 2-D convolutional layers is first adopted [37]. Then, researchers developed a set of CNN variants to achieve better classification performance [38], [39], [40], [41], [42], [43], [44], [45]. Some works attempt to boost classification performance via improving the network architecture. For example, Song et al. [38] introduce residual structures [39] to constitute a deep residual network, where features from different layers are fused for classification. Multiscale analysis is also introduced to construct multiscale CNN [40]. Considering that convolutional kernels significantly influence the performance of feature learning in CNN, some researchers try to modify the convolutional kernels. For instance, with the purpose of better capturing the characteristics of objects, dilated convolution and irregular convolution are, respectively, used in some recent works, i.e., [41], [42]. Besides, 3-D convolution-based CNNs are also developed [43] for HSIC. In addition, some researchers consider that patch-based CNN with single-branch architecture cannot make full use of spectral information of central pixel in the HSI patch. The reason is that the neighboring pixels within a convolution kernel may be from different classes and have very distinct spectrums, which corrupts the spectra characteristics. Therefore, two-branch CNNs are widely adopted in recent years [44], [45]. That is, an additional auxiliary branch, which inputs the spectrum of the central pixel into a subnetwork, is used to further extract discriminative spectral information. Compared with traditional methods, these CNN-based methods significantly improve the classification performance and demonstrate the effectiveness of CNN for HSIC. Nevertheless, the above-mentioned methods treat useful and useless information in the data equally regardless of whether the information is useful. It is likely that some useless information is overintroduced, which will result in a decrease in classification accuracy and a waste of computing resources.

More lately, the visual attention mechanism shows promising performance in enhancing image classification results, which has also been introduced as an effective way of feature optimization for spectral and spatial feature extraction processes [46], [47], [48]. In brief, this attention mechanism highlights the sensitivity to features containing the most valuable information, which can help learn discriminative features for more accurate classification. For instance, the attention-based adaptive spectral-spatial kernel improved residual network (A2S2K) [47] introduces the attention mechanism in the 3-D convolution-based residual module to effectively extract features. In addition, the residual spectral-spatial attention network (RSSAN) [48] embeds spectral-spatial attention modules in the residual-connection-based backbone network, enabling the network to learn spectral and spatial information adaptively. In addition, some studies based on topological relationships or graph information aggregation, and so on are also good to improve the utilization of local and nonlocal information of images [49], [50], [51]. Recently, self-attention-based Transformer networks have achieved remarkable success in the fields of natural language processing and computer vision, due to their ability to characterize long-term relationships. Several researchers have made some attempts to introduce Transformer networks into HSIC classification tasks [52], [53], [54], [55]. Yu et al. [52] used a pure transformer encoder (TE) for feature learning of serialized HSIs and demonstrated the feasibility of the Transformer structure for the HSI classification task. Yang et al. [53] performed global and local feature extraction on the input image based on 3-D convolution to obtain a better feature sequence for the Transformer network to learn. In [54], a masked autoencoder (MAE) and ViT-based architecture are used to pursue noise elimination to learn refined transformer features.

Most of the attention-based methods deal with all the spectral bands at one time to grasp the global representative features, while ignoring the spectral distribution information and spectral reflectance characteristics. That is, there is high spectral relevance between neighboring bands and low relevance across the long-range spectral bands; meanwhile, the spectral reflectance character in different bands may be sensitive to classify different materials. By the global attention mechanism, globally salient features will be learned well while some locally salient features maybe weaken, leading some important discrimination information that is useful to identify some classes lacking enough attention.

To overcome the above-mentioned limitations, we design a novel grouped multi-attention-driven DL network to enhance the feature discrimination capability for HSI classification. In specific, the proposed DL network is of two-branch architecture, including the pixelwise spectral feature learning branch and patchwise spatial–spectral feature learning branch. For the pixelwise branch, it takes the HSI pixel as the input and use several convolutional layers with the kernel size of 1×1 to enrich spectral features. The other one is the patchwise spectral-spatial feature extraction branch, which takes the HSI patch as the input to extract spectral-spatial features via a developed band-grouping guided multi-attention module.

To this end, the band-grouping mechanism, which adaptively partitions spectral bands into multiple groups, is first used in the patchwise spatial–spectral feature learning branch. Then, within each group, spectral and spatial attention modules are designed to weight the importance of input data, i.e., emphasizing the important spectral bands and spatial pixels for the extraction of spectral-spatial features. Finally, those patchwise spectral-spatial features are coupled with the pixelwise spectral features by the fully connected network and are next used to predict class labels by the Softmax function. We summarize the main contributions of the proposed method as follows.

1) The mechanism of adaptively grouping highly correlated adjacent bands is introduced in the DL-based HSI classification method for the first time. The motivation is inspired by the fact that discriminative information of certain material is generally included in one or several spectral ranges (e.g., green plants are sensitive to red and near-infrared bands). Rather than processing the whole bands, groupwise feature extraction can alleviate the influence of numerous unimportant bands.
2) We propose a novel multi-attention module for groupwise feature extraction, to make the network pay more attention to few critical and valuable spectral and spatial

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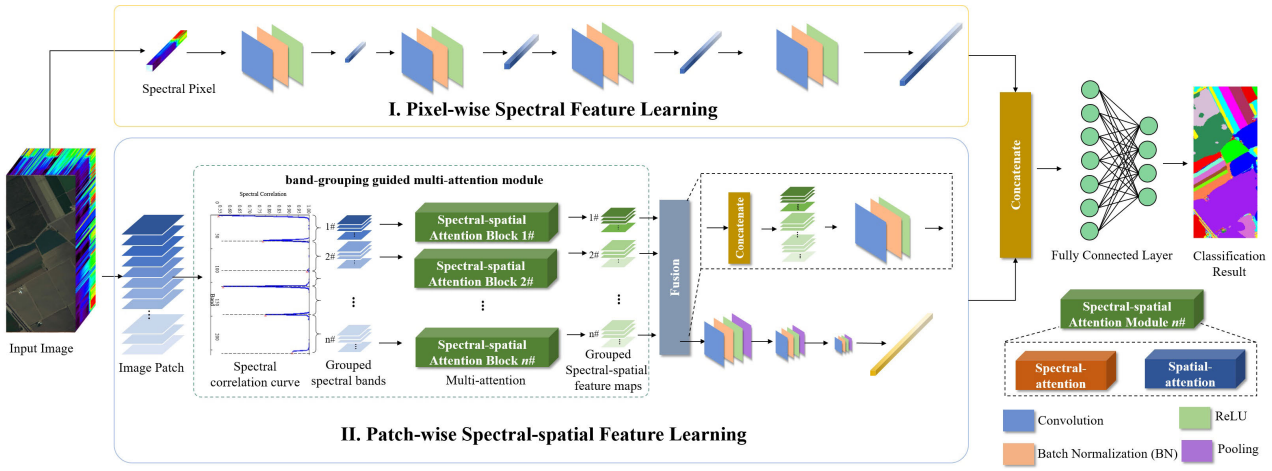


Fig. 1. Overview of the GMA-Net for HSIC.

information. To this end, the spectral attention and spatial attention blocks are, respectively, built. The former exploits the intragroup spectral correlations to generate spectral attention weights. The latter not only uses the intragroup salient information of pixels but also takes advantage of their intergroup communication, to better characterize the importance of spatial pixels.

3) A new two-branch network architecture is developed for HSIC in this work. One branch is responsible to learn highly discriminative spectral-spatial features using the CNN and multi-attention module. The other one provides complementary spectral information corrupted by the spatial convolutions. Features derived from two branches are fused to boost the final classification performance. Various experiments are conducted on well-known HSIs to confirm the effectiveness of the proposed method.

The rest of this article is organized as follows. Section II describes the proposed grouped multi-attention network (GMA-Net) in detail. The experimental study and results are discussed in Section III. Finally, this work is summarized in Section IV.

II. Methodology

A. Overview of the Proposed Method

Let $\boldsymbol{X} \in \mathbb{R}^{H\times L\times B}$ represent the HSI, where $H \times L$ and B, respectively, denote the spatial size and number of spectral bands. To better classify the HSI pixel $\boldsymbol{x}_i \in \mathbb{R}^{1\times 1\times B}$ with spectral and spatial information, the HSI patch $\boldsymbol{X}_i \in \mathbb{R}^{s\times s\times B}$ is usually cropped from $\boldsymbol{X}$ and input into the CNN to extract spectral-spatial features. Here, $s \times s$ is the patch size, and $\boldsymbol{x}_i$ is the central pixel of $\boldsymbol{X}_i$. As known, the spatial convolution operation in the CNN will aggregate the information of adjacent pixels, which leads to the degradation of spectral characteristics as the adjacent pixels are from different categories. Therefore, in addition to the HSI patch, the pure spectral information of the HSI pixel is also required. To this end, we input both the HSI pixel $\boldsymbol{x}_i$ and the HSI patch $\boldsymbol{X}_i$ into an uniform network, which is named GMA-Net and owns a two-branch architecture. The overview of the GMA-Net is demonstrated in Fig. 1.

Specifically, the pixelwise spectral feature extraction branch (called pixel branch for short) takes pixel $\boldsymbol{x}_i$ as the input to extract the spectral feature $\boldsymbol{O}_{\text{pixel}}$. The patchwise spectral-spatial feature extraction branch (called patch branch for short) takes HSI patch $\boldsymbol{X}_i$ as the input to extract the spectral-spatial feature $\boldsymbol{O}_{\text{patch}}$.

Finally, the output features by these two branches are concatenated and injected into the output layer constituted by the fully connected layer and the softmax function. The label vector $\boldsymbol{y}$ (for convenience, we will ignore the superscript) by the following equation:

$$\boldsymbol{y} = \text{Softmax}(f_{\text{FC}}(C(\boldsymbol{O}_{\text{pixel}}, \boldsymbol{O}_{\text{patch}}))) \tag{1}$$

where $C(\cdot)$ denotes the operation of concatenating multiple features, $f_{\text{FC}}(\cdot)$ means feature learning by the fully connected layer, and $\boldsymbol{y}$ records the classification probability with respect to different categories.

B. Pixelwise Spectral Feature Learning

In the pixel branch, a simplified CNN is designed to extract spectral features from input $\boldsymbol{x}_i$. Considering that the pixel $\boldsymbol{x}_i$ is a 1-D spectral vector, we adopt the 1×1 convolution kernels, together with batch normalization (BN) after each convolution before activation to speed up the convergence of the CNN. As a consequence, the CNN constituted by four cascaded convolutional layers with similar structures can be represented by the following equation:

$$\begin{aligned}
\boldsymbol{O}_{\text{pixel}} &= \sigma(f_{\text{BN}}(\boldsymbol{W}_4 * \boldsymbol{O}_3 + \boldsymbol{b}_4)) \\
\boldsymbol{O}_3 &= \sigma(f_{\text{BN}}(\boldsymbol{W}_3 * \boldsymbol{O}_2 + \boldsymbol{b}_3)) \\
\boldsymbol{O}_2 &= \sigma(f_{\text{BN}}(\boldsymbol{W}_2 * \boldsymbol{O}_1 + \boldsymbol{b}_2)) \\
\boldsymbol{O}_1 &= \sigma(f_{\text{BN}}(\boldsymbol{W}_1 * \boldsymbol{x}_i + \boldsymbol{b}_1))
\end{aligned} \tag{2}$$

where $\boldsymbol{O}_{\text{pixel}}$ denotes the output of the pixel branch, $\boldsymbol{O}_1$, $\boldsymbol{O}_2$, and $\boldsymbol{O}_3$ are features acquired by the corresponding layers, and $\{\boldsymbol{W}_1, \ldots, \boldsymbol{W}_4\}$ and $\{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_4\}$ are, respectively, the weights and bias of different layers. We drop the bias in Sections II-D and II-E to simplify the description of the proposed method.

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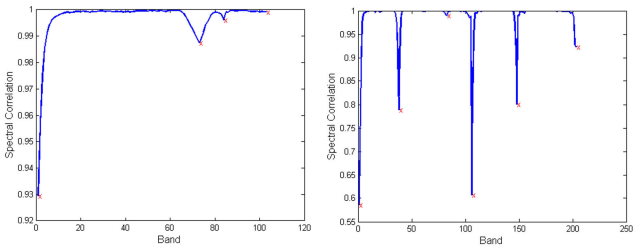


Fig. 2. Spectral correlation coefficient curves. (a) UP image. (b) SA image.

In addition, $*$ means the convolution operation, $f_{\mathrm{BN}}(\cdot)$ denotes the BN operation, and $\sigma(\cdot)$ represents the activation function. Here, the linear rectification function (ReLU) activation function is adopted. The number of convolution kernels used for these four convolutional layers is successively set to 32, 64, 128, and 256.

C. Patchwise Spectral-Spatial Feature Learning

According to the spectral reflection characteristic that adjacent spectral bands and spatial pixels tend to be similar, useful information is generally surrounded by spectral and spatial redundancy, and thus requires special attention. Furthermore, the high-dimensional data make it difficult to focus on the local fine discrimination. In the patch branch, to solve the above problems, we develop a band-grouping guided multi-attention module to promote the performance of spectral-spatial feature learning. As shown in Fig. 1, we first adaptively divide the spectral bands into several low-dimensional band groups. Based on this, we design the multi-attention framework to emphasize the useful information and suppress useless information of multiple groups with the learned attention weights. After that, the attention guided features are delivered to a CNN for further refinement.

To implement the division of all the spectral bands, the adaptive band-grouping method is developed based on the spectral correlation analysis. As shown in Fig. 2 where the correlation coefficients of adjacent spectral bands are plotted, there are some remarkable valley points. These valley points indicate that spectral bands are piecewise similar. Therefore, as long as we can find these valley points, adaptive band-grouping is achieved. To this end, the Pearson correlation coefficients ρ of adjacent spectral bands are first calculated by the following equation:

$$\rho(j) = \frac{\mathrm{Cov}(\boldsymbol{y}_j, \boldsymbol{y}_{j+1})}{\sigma_j \sigma_{j+1}} \tag{3}$$

where $\boldsymbol{y}_j$ and $\boldsymbol{y}_{j+1}$, respectively, represent the jth and $j+1$th spectral bands, $\mathrm{Cov}(\cdot)$ means the calculation of covariance, and σ_j and σ_{j+1} are the standard deviation of $\boldsymbol{y}_j$ and $\boldsymbol{y}_{j+1}$, respectively. Then, the points which are smaller than some adjacent points are regarded as the valley points. The processing can be formulated as follows:

$$\mathrm{Index}(j) = \begin{cases} 1, & \rho(j) < \min(\bar{\rho}_l(j), \bar{\rho}_r(j)) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

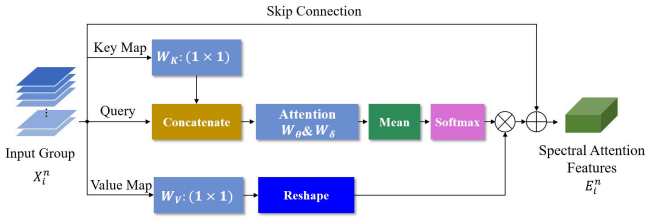


Fig. 3. Detailed structure of the spectral attention block.

where $\mathrm{Index}(j) = 1$ indicates the valley point, and $\bar{\rho}_l(j) = 1/M \sum_{m \in \Omega_l} \rho_m$ and $\bar{\rho}_r(j) = 1/M \sum_{m \in \Omega_r} \rho_m$ are the average values of correlation coefficients with respect to M adjacent points, respectively, located in the right and left sides of point j (M is fixed to 4 for different HSIs). By means of these valley points, spectral bands are adaptively grouped. Finally, the postprocessing that merges adjacent groups with high correlation or very few bands is used to further refine and output n band groups $\{\boldsymbol{X}_i^1, \boldsymbol{X}_i^2, \ldots, \boldsymbol{X}_i^n\}$.

For these band groups, the multi-attention framework is developed based on intragroup correlation calculation and intergroup collaboration, to learn attention guided features. The structure of this model can be seen in Fig. 1. First, a spectral module and a spatial module can make use of the intragroup information to, respectively, acquire the spectral and spatial attention weights for each group. The detailed description of spectral and spatial modules is demonstrated in the following sections.

Since the material has different spectral reflection characteristics for very different wavelengths, various groups tend to have distinct attention information. Then, based on this intergroup relationship, we use the attention information of different groups to jointly optimize features. Specifically, let $\boldsymbol{F}_n$ denote the attention maps of the nth group obtained by the spatial attention block. The squeezed attention information $1-\bar{\boldsymbol{F}}_n$ can be useful for other groups, where $\bar{\boldsymbol{F}}_n$ is the averaged map of $\boldsymbol{F}_n$. Therefore, the attention guided spatial features $\boldsymbol{Z}_i^n$ with respect to the nth group $\boldsymbol{X}_i^n$ can be obtained by the following equation:

$$\boldsymbol{Z}_i^n = C\left(\boldsymbol{F}_n \boldsymbol{X}_i^n, R(1-\bar{\boldsymbol{F}}_n)\boldsymbol{X}_i^2, \ldots, R(1-\bar{\boldsymbol{F}}_n)\boldsymbol{X}_i^{n-1}\right) \tag{5}$$

where $1-\bar{\boldsymbol{F}}_n$ is reshaped to the size of various groups by $R(\cdot)$. In addition, the attention guided spectral features $\{\boldsymbol{E}_i^1, \ldots, \boldsymbol{E}_i^n\}$ of total n groups are achieved by the spectral attention block.

Next, the grouped spectral-spatial features, achieved by concatenating $\{\boldsymbol{Z}_i^1, \ldots, \boldsymbol{Z}_i^n\}$ and $\{\boldsymbol{E}_i^1, \ldots, \boldsymbol{E}_i^n\}$, are fused to further refine the emphasized spectral-spatial information via a convolutional layer. The convolutional layer adopts 1×1 convolutional kernels for the feature fusion. Finally, the fused features are delivered to a CNN which consists of three convolutional layers (kernel size is set to 5×5), to acquire spectral-spatial feature $\boldsymbol{O}_{\mathrm{patch}}$.

D. Spectral Attention Block

For each spectral group, the spectral attention block aims at emphasizing the important spectral bands that contribute to feature learning and final classification. To achieve this

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purpose, a new learnable spectral attention block, which has the detailed structure shown in Fig. 3, is designed based on the self-attention mechanism. First, we achieve the query $\boldsymbol{Q}$, key $\boldsymbol{K}$, and value $\boldsymbol{V}$ by mapping the input group $\boldsymbol{X}_i^n$ to different feature spaces. That is, $\boldsymbol{K} = \boldsymbol{W}_K * \boldsymbol{X}_i^n$, $\boldsymbol{V} = \boldsymbol{W}_V * \boldsymbol{X}_i^n$, and $\boldsymbol{Q} = \boldsymbol{X}_i^n$, where $\boldsymbol{W}_K$ and $\boldsymbol{W}_V$ are the embedding matrixes implemented by the convolution operation. By means of 1×1 convolutional kernels, the mapping by the embedding matrices can effectively preserve the spectral information in Q and V. Then, the spectral attention maps $\boldsymbol{F}_n'$ of the nth group are generated by collaboratively using K and Q, which can be formulated as follows:

$$\boldsymbol{F}_n' = \text{Softmax}(\boldsymbol{W}_\delta * \boldsymbol{W}_\theta * C(\boldsymbol{K}, \boldsymbol{Q})) \tag{6}$$

where $\boldsymbol{W}_\theta$ is the convolutional layer with 1×1 convolutional kernels and the ReLU activation function, and $\boldsymbol{W}_\delta$ is the 1×1 convolutional layer followed by the mean operation and the Softmax function. Based on this, the critical spectral information with respect to $\boldsymbol{X}_i^n$ is emphasized by multiplying $\boldsymbol{V}$ and the attention maps, i.e., $\boldsymbol{P}_i^n = \boldsymbol{F}_n' \boldsymbol{V}$.

Finally, inspired by the residual network, we use the skip connection to preserve more useful information, and thus obtain the spectral attention features $\boldsymbol{E}_i^n$ of the nth group by the following equation:

$$\boldsymbol{E}_i^n = \lambda_1 \boldsymbol{P}_i^n + \lambda_2 \boldsymbol{X}_i^n \tag{7}$$

where λ_1 and λ_2 are the learnable weights used to tradeoff the contribution of $\boldsymbol{P}_i^n$ and $\boldsymbol{X}_i^n$. All the spectral attention features $\{\boldsymbol{E}_i^n\}$ will be stacked and used for the following feature fusion and feature learning in the CNN.

E. Spatial Attention Block

For each spectral group, the spatial attention block aims at generating the spatial attention maps which consist of attention weights used to emphasize the salient spatial information and contribute to feature learning and final classification. To implement this purpose, we designed a learnable spatial attention block, which is demonstrated in Fig. 4. The downsampling is first applied to the input group $\boldsymbol{X}_i^n$ to reduce redundancy and highlight salient information. Instead of using the pooling operation without learning, we adopt a learnable convolutional layer for downsampling. Specifically, the 3-D convolution and nearest interpolation are used, which can be formulated as follows:

$$\boldsymbol{F}_t = \text{upsampling}\left(\boldsymbol{W}_\alpha * \boldsymbol{X}_i^n\right) \tag{8}$$

where $\boldsymbol{F}_t$ denotes the feature maps used to roughly extract salient spatial information, $\boldsymbol{W}_\alpha$ denotes a 3-D convolution layer with the kernel size of $3 \times 3 \times 1$, and upsampling$(\cdot)$ is the nearest interpolation-based upsampling operation to recover the spatial sizes of feature maps.

After that, $\boldsymbol{F}_t$ is used for further feature learning by means of a 3-D convolutional layer. Moreover, the learned features are normalized by the Sigmoid function to generate the attention weights. As a result, the attention maps $\boldsymbol{F}_n$ can be acquired by the following equation:

$$\boldsymbol{F}_n = \sigma(f_{\text{BN}}(\boldsymbol{W}_\beta * F_t)) \tag{9}$$

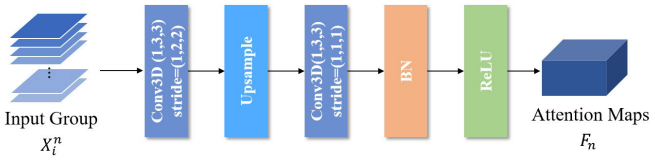


Fig. 4. Detailed structure of the spatial attention block.

where $\boldsymbol{W}_\beta$ denotes the 3-D convolution with kernel size $3 \times 3 \times 1$, and σ means the activation function (the Sigmoid function is used here). Finally, $\boldsymbol{F}_n$ is substituted into (5) to obtain the attention guided features $\boldsymbol{Z}_i^n$.

III. Experiments and Analysis

In this section, the three public HSIs used in the experiments are first described. Then, the evaluation metrics, the experimental environment, and the settings of some important parameters are described in detail. Next, the proposed GMA-Net is compared with several state-of-the-art methods by means of widely used evaluation metrics and visualization results, to evaluate the classification performance. After that, the effects of single and multiple components of the proposed GMA-Net are analyzed in ablation studies. Finally, some critical parameters affecting the performance of the proposed model are discussed, and the computational complexity is analyzed.

A. Datasets' Description

To test the effectiveness of the proposed method, we conducted experiments on three popular HSIs, including University of Pavia (UP), Salinas (SA), and Houston (HO). The false color composite and the ground truth are, respectively, shown in Figs. 5–7. Detailed descriptions of the three images are given as below.

1) University of Pavia: This image is acquired by the reflection optical system imaging spectrometer (ROSIS), over the scene of campus in the UP. The size of image is $610 \times 340 \times 103$ pixels, with the spatial resolution being 1.3 m/pixel. This image covers the spectral range of 0.43–0.86 μm with a spectral resolution of 3.74 nm.

2) Salinas: This image is collected by the airborne visible–infrared imaging spectrometer (AVIRIS), over the scene of agriculture in SA Valley, California, USA. The size of SA is $512 \times 217 \times 204$ with a spatial resolution of 3.7 m/pixel. The corresponding wavelength range of the SA images is 0.4–2.5 μm.

3) Houston: This image is released in the 2013 IEEE geoscience and remote sensing society (GRSS) data fusion competition and covers the complex scene of the University of Houston and the adjacent downtown area. Its size is $1905 \times 349 \times 114$ with a spatial resolution of 2.5 m/pixel. The wavelength range is from 0.364 to 1.046 μm.

B. Experimental Setup

1) Evaluation Metrics: To evaluate the performance of the proposed GMA-Net, several widely used metrics are used for evaluation, including the class accuracy (CA),

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TABLE I

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UP IMAGE. THE OPTIMAL VALUES FOR EACH CASE ARE GIVEN IN BOLD

Class	Name	Train	Test	SSRN [43]	RSSAN [48]	A2S2K [47]	MTGAN [36]	SPRN [46]	GMA-Net
1	Asphalt	20	6611	75.05±26.76	77.09±4.44	90.27±6.06	95.59±2.46	86.02±6.20	**96.72±2.09**
2	Meadows	20	18629	87.88±7.87	93.64±2.57	97.77±1.31	98.47±1.49	97.25±2.33	**99.66±0.32**
3	Gravel	20	2079	55.51±22.26	62.05±9.91	91.93±3.79	93.96±4.31	91.24±6.74	**99.41±0.66**
4	Trees	20	3044	95.84±3.33	94.01±1.78	**97.11±1.93**	95.13±2.11	93.80±3.16	97.07±1.62
5	Metal sheets	20	1325	99.97±0.05	98.28±2.67	99.54±1.25	99.42±0.66	99.92±0.15	**100.00±0.00**
6	Bare soil	20	5009	84.71±14.91	74.79±9.64	90.37±4.90	98.31±3.30	94.41±2.49	**98.58±1.30**
7	Bitumen	20	1310	95.62±6.20	86.95±5.88	96.52±6.64	97.80±2.67	99.69±0.22	**99.99±0.03**
8	Bricks	20	3662	92.27±10.64	74.25±6.28	93.03±1.93	95.36±2.55	92.09±5.73	**98.26±0.91**
9	Shadows	20	927	99.88±0.23	98.34±1.47	99.83±0.18	96.05±2.05	99.41±0.78	**99.88±0.13**
OA	-	-	-	85.76±6.30	85.70±2.10	95.06±1.00	97.23±0.77	94.40±1.56	**98.78±0.37**
AA	-	-	-	87.42±4.40	84.37±2.42	95.15±0.99	96.68±0.62	94.87±1.15	**98.84±0.36**
κ	-	-	-	81.59±7.63	82.07±4.80	93.45±1.32	96.34±1.01	92.60±2.03	**98.39±0.49**

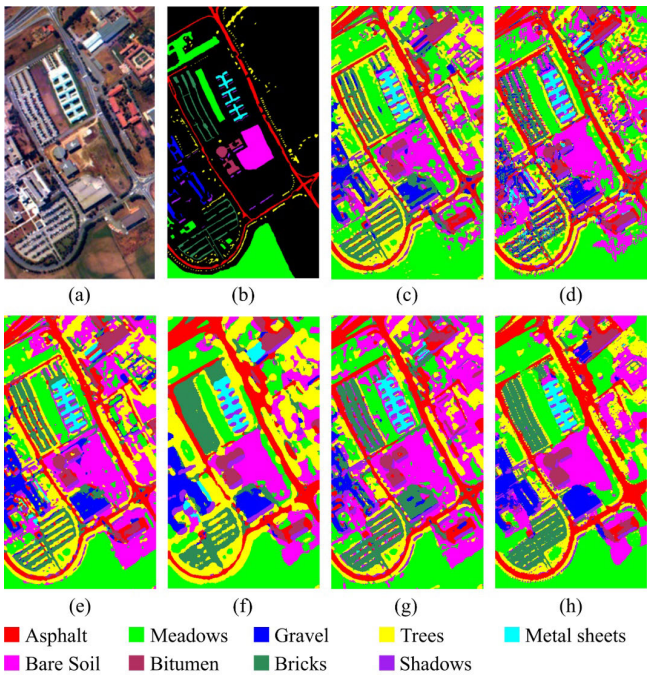


Fig. 5. Classification maps for the UP image. (a) False composite map. (b) Ground-truth map. (c)–(h) Classification maps of SSRN, RSSAN, A2S2K, MTGAN, SPRN, and our GMA-Net.

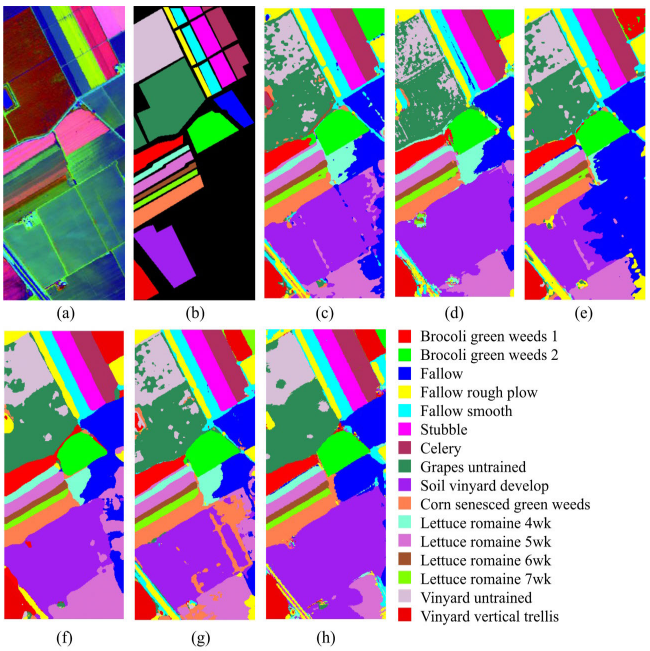


Fig. 6. Classification maps for the SA image. (a) False composite map. (b) Ground-truth map. (c)–(h) Classification maps of SSRN, RSSAN, A2S2K, MTGAN, SPRN, and our GMA-Net.

overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). CA indicates the ratio between the number of correctly classified and the total number of test samples in each class. AA is the average of CA from all the classes. OA represents the ratio between the number of all correctly classified samples and the number of overall test samples. Kappa coefficient can measure the relationship between the possibility of expected agreement and disagreement and is a more robust accuracy assessment than OA using the confusion matrix.

2) Configuration Detail: In the experiments, we randomly select a small amount of labeled samples from each class to constitute training dataset and use the remainder for testing. To avoid the influence of random operations, we repeat each experiment ten times and report the average results and standard deviations of various metrics. The proposed GMA-Net and other networks used for comparison in this article are implemented on the PyTorch platform. All the networks are run on a computer which has an Intel(R) Core(TM) i9-10900K CPU, 32-GB RAM, NVIDIA GeForce RTX 3090 GPU, and the software environment of 64-bit Windows 10 operating system.

3) Parameter Setting: For the DL based classification methods used for comparison, we run their codes released on the web and do our best to achieve the highest classification accuracies for three HSIs. Some of their hyperparameters are set to default values reported as suggested in corresponding papers. For the proposed method, we use the Adam Optimizer to optimize the network parameters with a batch size of 32. The number of training iterations is fixed to 200 epochs. Two important parameters, i.e., the learning rate and the patch size, are set to 0.0001 and 31×31, respectively.

C. Comparison With the State-of-the-Art Methods

To validate the effectiveness of the proposed method, we compare the GMA-Net with several state-of-the-art HSI

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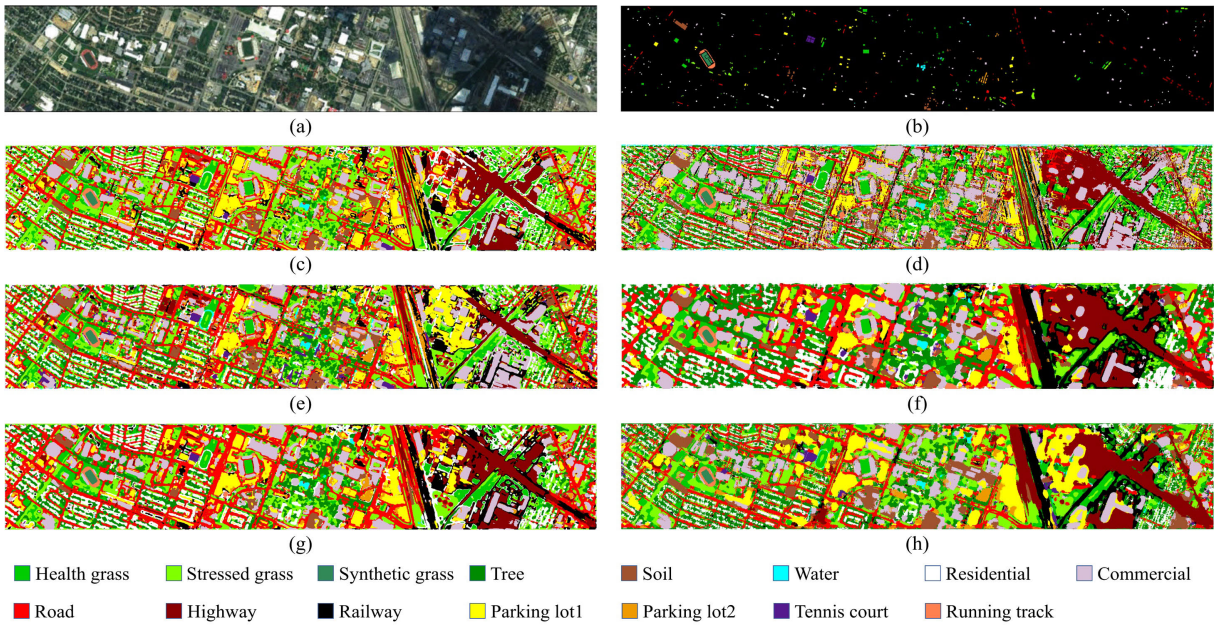


Fig. 7. Classification maps for the HO image. (a) False composite map. (b) Ground-truth map. (c)–(h) Classification maps of SSRN, RSSAN, A2S2K, MTGAN, SPRN, and our GMA-Net.

classification methods, i.e., the spectral-spatial residual network (SSRN) [43], RSSAN [48], A2S2K improved residual network [47], multitask generative adversarial network (MTGAN) [36], and spectral partitioning residual network (SPRN) [46]. Among the compared methods, MTGAN is a representative non-CNN method which acquires extremely high classification accuracy, while the rest of the methods adopt 2-D CNN or 3-D CNN to extract spectral-spatial features, and SSRN is a classical CNN variant for HSI classification. Besides, RSSAN, A2S2K, and SPRN preliminarily introduce the attention mechanism to HSI classification and are related to the proposed GMA-Net. Based on the above considerations, we conduct comparison experiments on three HSIs. To make a fair comparison, we use the same training and testing data for different methods. After that, the quantitative and qualitative results with respect to three HSIs are as demonstrated below.

1) University of Pavia: The UP is the first HSI used to evaluate the performance of different classification methods. As can be seen in Table I, we only select 20 labeled samples per class for training and use the reminder for testing.

The quantitative classification results (i.e., the values of CA, OA, AA, and κ) are shown in Table I. It can be seen that the SSRN and RSSAN methods are not well-trained with such limited training samples, and thus acquire much lower classification accuracy than other methods, e.g., their OAs are much less than 90%. Due to the better utilization of attention mechanism, A2S2K and SPRN show comparable classification performance, e.g., their OAs can reach to about 95%. Even A2S2K achieves the optimal CA on the "Trees" category. Compared with the above four methods, the MTGAN introduces the game mechanism and exploits unsupervised information to better train the network. As a result, the main metrics of the MTGAN are obviously higher than those of four CNN-based methods. Thanks to the use of developed grouped multi-attention and two-branch architecture, the proposed GMA-Net can significantly improve the classification performance of CNN, and thus achieve the optimal values in most cases, especially for OA, AA, and κ (the improvement can reach to about 2%). Although the CA of our method on the "Trees" category is not optimal, the gap between GMA-Net and A2S2K is really limited, i.e., only 0.04%. Moreover, the standard deviation 1.62% of GMA-Net is less than 1.93% of A2S2K on this category, which means that GMA-Net is more robust than A2S2K for classification. Based on these results, we can draw the preliminary conclusion that the proposed GMA-Net is superior to the compared methods in this HSI.

Besides, the visual classification results on the UP image are shown in Fig. 5. It can be observed that the classification maps of CNN-based methods (i.e., SSRN, RSSAN, A2S2K, and SPRN) have some pixelwise or regionwise noises, especially in areas constituted by unlabeled pixels. Although these methods simultaneously exploit spectral and spatial information, their models may suffer from the problem of overfitting since limited training samples are given. As a consequence, the inference ability of their models is restricted, especially for unlabeled samples which may contain unseen spectral-spatial structures. Although MTGAN achieves higher OA, AA, and κ than these CNN-based methods, it generates an oversmooth classification map as shown in Fig. 5(f), especially for the "Trees" category marked with yellow color. Compared with these methods, the proposed GMA-Net has the most accurate classification map. Furthermore, much less noises are represented without oversmooth areas in the classification map, and the boundaries of different objects are clear and sharp. Based on these observations, it can be concluded that our method performs better classification results than other classification method even in the unlabeled areas.

2) Salinas and Houston: For these two HSIs, we also select 20 labeled samples per class to constitute training data

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TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SA IMAGE. THE OPTIMAL VALUES FOR EACH CASE ARE GIVEN IN BOLD

Class	Name	Train	Test	SSRN [43]	RSSAN [48]	A2S2K [47]	MTGAN [36]	SPRN [46]	GMA-Net
1	Brocoli green weeds 1	20	1989	98.04±2.77	99.95±0.07	99.20±1.14	**100.00±0.00**	99.98±0.04	**100.00±0.00**
2	Brocoli green weeds 2	20	3706	**100.00±0.00**	99.61±0.10	99.95±0.08	99.34±0.48	**100.00±0.00**	**100.00±0.00**
3	Fallow	20	1956	99.29±1.01	98.95±1.48	**99.95±0.07**	97.63±3.36	98.37±1.73	**99.95±0.07**
4	Fallow rough plow	20	1374	98.47±1.85	98.22±0.26	99.67±0.47	99.02±1.39	99.57±0.62	**99.71±0.41**
5	Fallow smooth	20	2658	88.68±14.10	98.91±1.44	99.14±0.95	97.52±3.40	96.82±1.63	**99.31±0.29**
6	Stubble	20	3939	99.77±0.33	99.98±0.04	99.97±0.00	99.86±0.13	**100.00±0.00**	99.91±0.08
7	Celery	20	3559	99.97±0.00	99.61±0.52	99.80±0.20	**99.99±0.02**	99.78±0.16	**99.99±0.02**
8	Grapes untrained	20	11251	84.10±5.83	90.65±4.11	87.81±5.20	**91.60±2.76**	87.17±2.16	86.15±1.86
9	Soil vinyard develop	20	6183	99.52±0.01	99.96±0.06	99.78±0.32	**100.00±0.00**	99.84±0.18	**100.00±0.00**
10	Corn senesced green weeds	20	3258	94.88±3.64	98.30±0.85	96.09±0.98	99.25±0.49	**99.42±0.18**	97.85±1.30
11	Lettuce romaine 4wk	20	1048	98.95±1.34	98.48±0.81	98.67±1.21	99.10±0.06	98.57±0.81	**99.67±0.21**
12	Lettuce romaine 5wk	20	1907	**100.00±0.00**	99.77±0.33	99.95±0.07	99.56±0.63	**100.00±0.00**	99.98±0.04
13	Lettuce romaine 6wk	20	896	99.61±0.55	99.33±0.95	**99.89±0.16**	99.72±0.40	98.05±2.76	99.78±0.02
14	Lettuce romaine 7wk	20	1050	95.96±3.30	96.76±4.17	90.95±9.29	97.67±2.89	97.72±2.69	**98.57±0.81**
15	Vinyard untrained	20	7248	73.97±5.57	78.87±9.02	77.55±7.40	83.49±2.50	79.33±2.55	**97.04±3.87**
16	Vinyard vertical trellis	20	1787	96.00±3.76	98.94±1.51	97.23±2.02	98.94±1.51	97.32±0.56	**99.22±1.03**
OA	-	-	-	91.85±1.07	94.75±2.36	93.77±0.08	95.56±1.22	94.04±0.06	**96.46±0.08**
AA	-	-	-	95.45±0.61	97.27±1.38	96.60±0.18	97.67±0.81	96.99±0.17	**98.57±0.14**
κ	-	-	-	90.93±1.18	94.15±2.64	93.05±0.07	95.06±1.35	93.36±0.06	**96.06±0.04**

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HO IMAGE. THE OPTIMAL VALUES FOR EACH CASE ARE GIVEN IN BOLD

Class	Name	Train	Test	SSRN [43]	RSSAN [48]	A2S2K [47]	MTGAN [36]	SPRN [46]	GMA-Net
1	Grass_healthy	20	1231	96.65±4.88	95.09±2.84	98.11±1.53	90.32±7.54	**98.31±2.79**	93.32±5.83
2	Grass_stressed	20	1234	91.56±4.64	93.58±5.35	**98.54±1.36**	89.05±5.75	92.81±4.74	89.57±6.74
3	Grass_synthetic	20	677	99.91±0.22	98.63±3.69	99.94±0.13	99.54±0.46	99.98±0.05	**99.99±0.05**
4	Tree	20	1224	94.00±4.12	92.09±4.01	94.75±2.71	93.38±2.64	91.65±3.60	**96.17±3.06**
5	Soil	20	1222	98.46±2.85	94.35±3.17	98.91±1.75	98.98±1.17	**99.29±1.12**	99.25±1.31
6	Water	20	305	94.72±5.37	88.82±3.91	**96.79±4.86**	93.21±6.65	96.17±5.57	95.31±5.12
7	Residential	20	1248	78.32±14.77	79.69±5.22	82.04±7.76	**90.77±2.92**	72.42±10.48	90.32±3.60
8	Commercial	20	1224	73.44±5.55	69.20±3.50	**73.64±6.21**	73.53±8.44	67.55±7.64	72.82±6.59
9	Road	20	1232	77.84±6.61	80.37±6.57	78.95±5.12	86.36±2.59	84.70±4.81	**92.09±3.59**
10	Highway	20	1207	66.94±15.09	90.45±5.03	87.36±6.78	84.67±11.13	86.58±9.13	**98.45±1.91**
	Railway	20	1215	78.96±7.09	72.61±7.60	83.00±8.56	91.82±2.94	85.09±7.07	**93.23±4.44**
12	Parking_lot1	20	1213	90.03±6.59	87.17±4.92	91.21±4.35	**94.30±3.22**	83.98±8.39	93.68±3.67
13	Parking_lot2	20	449	93.99±1.33	78.87±6.99	**95.46±1.38**	95.24±3.04	83.84±3.80	91.47±5.44
14	Tennis_court	20	408	99.95±0.15	98.21±2.07	99.71±0.68	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
15	Running_track	20	640	99.88±0.26	96.50±3.16	99.86±0.33	97.91±1.37	99.38±1.42	**100.00±0.00**
OA	-	-	-	86.92±1.04	86.77±1.66	90.36±0.98	90.72±1.18	87.92±1.34	**92.88±0.64**
AA	-	-	-	88.98±0.89	87.71±1.40	91.90±0.84	91.94±1.21	89.43±1.12	**93.71±0.75**
κ	-	-	-	85.85±1.12	86.69±3.59	89.57±1.06	89.97±1.28	86.94±1.45	**92.30±0.69**

for various methods. The obtained values of CAs, OA, AA, and κ by different methods on the SA and HO images are, respectively, reported in Tables II and III. The visual classification maps of different methods are, respectively, shown in Figs. 6 and 7.

According to Table II, SSRN has the lowest classification accuracy (OA, AA, and κ) while three attention-based methods, i.e., A2S2K, RSSAN, and SPRN, have similar classification performance. Besides, MTGAN outperforms the above four methods with the improvement of more than 1% with respect to OA. Compared with these methods, the proposed GMA-Net achieves the optimal OA, AA, κ, as well as optimal CAs in most classes (11 classes). For three of the other five categories, accuracies of the GMA-Net are approximate to 99.9%, and thus are comparable to the optimal values.

Different from the UP and SA images, the HO image contains more complex urban scenes. Therefore, as can be seen in Table III, the existing classification methods achieve comparatively low classification accuracy. That is, the highest OA is 90.72% while the OA acquired by three of these methods is about 87%. When compared with these methods, our method significantly improves the classification accuracy (OA) from 90.72% to 92.88%. The similar situation occurs in the metrics of AA and κ.

According to the visual results shown in Figs. 6 and 7, the GMA-Net has an accurate classification map with less "noises" and clear boundaries, which indicates that the proposed method is comparable to the existing state-of-the-art methods. The above quantitative and qualitative comparison validates the effectiveness of the proposed GMA-Net.

D. Ablation Study

The core of GMA-Net is the proposed grouped multi-attention which mainly contains three important components, i.e., the band-grouping, spectral attention block, and spatial attention block. Besides, the pixelwise spectral feature learning branch is constructed to further improve the classification accuracy and can be regarded as another important component of GMA-Net. Therefore, we conduct experiments to verify the effectiveness of these four components.

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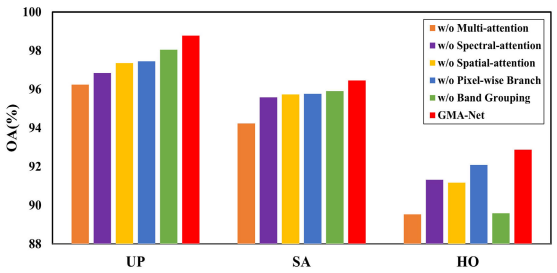


Fig. 8. Effect of adaptive band-grouping, pixelwise branch, and different attention modules on GMA-Net on three images.

First, we remove one component from the GMA-Net in turn and, respectively, use "w/o band grouping," "w/o spectral-attention," "w/o spatial-attention," "w/o multi-attention," and "w/o pixelwise branch" to represent different versions of GMA-Net. In addition, we also care about that multiple components may play the same role in promoting classification performance of the GMA-Net, especially for relevant components. Therefore, multiple relevant components, i.e., the band-grouping, spectral attention module, and spatial attention module, are simultaneously removed from GMA-Net, which is denoted as "w/o multi-attention." Then, these methods are, respectively, used to classify the UP, SA, and HO images. Finally, the key metric (OA) is used to evaluate the classification accuracy of different methods. The detailed OA values in terms of three HSIs are used to plot a histogram and shown in Fig. 8. The bars with different colors in Fig. 8 denote different versions of GMA-Net. Red bars represent the OA of the original GMA-Net while bars with other colors indicate the OA of variants after removing one or two components. Obviously, the red bars are much higher than other bars. It means that after removing one component or multiple components from GMA-Net, the OA values dramatically decline, especially when multiple components are removed (i.e., orange bars are always the lowest for different HSIs). Furthermore, the three HSIs cover different types of scenes, which makes different components play a different role in GMA-Net. For example, the height of the green bar is very close to those of the red bar for the UP image, but the opposite results are achieved for the HO image. Basically, it can be concluded that these four components are definitely essential to improve the performance of GMA-Net.

E. Parameter Discussion

1) Learning Rate and Patch Size: In the GMA-Net, the learning rate determines the step of gradient descent and plays an important role in the convergent speed and convergence of network training. The patch size determines how much spatial information is used for spectral-spatial feature extraction. Moreover, different patch sizes means different lengths of input signals which may require certain learning rates. Therefore, both the learning rate and patch size are relevant and act as important parameters of GMA-Net. They are discussed here.

We use grid search to find the optimal learning rate and patch size of the proposed method for the UP, SA, and HO images. Unlike the general approach of adjusting only one parameter with other fixed parameters, we adjust two parameters together to more accurately find the optimal values. To be specific, the learning rate is, respectively, set to 0.01, 0.003, 0.001, 0.0003, 0.0001, and 0.00003. In the meantime, we adopt different patch sizes, i.e., 19×19, 23×23, 27×27, 31×31, and 35×35. For the three HSIs, the values of OA acquired by GMA-Net under different learning rates and patch sizes are obtained. We use those values to draw three 3-D parametrization diagrams and, respectively, demonstrate them in Fig. 9(a)–(c).

As shown in Fig. 9, for different HSIs, the classification accuracy of the GMA-Net gradually increases to the highest values as the learning rate decreases and the patch size increases, and then gradually drops for much smaller learning rate (e.g., 0.00003) and much larger patch size (e.g., 35). Basically, the highest OAs are achieved when the learning rate is 0.0001 and the patch size is 31×31. Thanks to the use of attention mechanism, our method can adopt relatively large patch sizes (e.g., 31×31) since attention mechanism contributes to adaptively extract useful information from much spectral-spatial information. Therefore, when the patch is very small (e.g., 19×19), it may contain too little useful information to better learn features for classification. On the contrary, too large patch will contain more complex structures and more pixels from different categories, which corrupt the performance of feature leaning for classification. Based on the above results and analysis, it is reasonable to, respectively, set the learning rate and patch size to 0.0001 and 31×31.

2) Number of Training Samples: To further verify the robustness of the proposed GMA-Net with respect to different numbers of training samples, we conduct experiments by changing the number of selected samples from each class (denoted by N). Specifically, the number is changed from 5 to 30 with the step-size of 5, for the three HSIs. The corresponding classification accuracy (OA) of various methods on the three HSIs is used to plot three line charts which are shown in Fig. 10. It can be observed that most of the curves from different methods present a rise trend along with the increase in N, as networks are better trained with more training samples. There exist some fluctuations for the curves of several methods, such as SSRN and A2S2K. Especially for SSRN, the curve decreases dramatically when $N = 20$, which is mainly caused by random initialization of the network. On the other hand, when compared with other methods, the SSRN method is not such robust for random initialization. Furthermore, for the three HSIs, the curves of GMA-Net are always beyond those of other methods, which means the proposed method performs the best on different N.

F. Computational Complexity Analysis

We evaluate the complexity and computational cost of different models with respect to the number of trainable weight parameters updated during the back propagation procedure, the million floating-point operations (MFLOPs), and the training and testing time. All the experiments are conducted on the Pavia dataset with a computing server equipped with an

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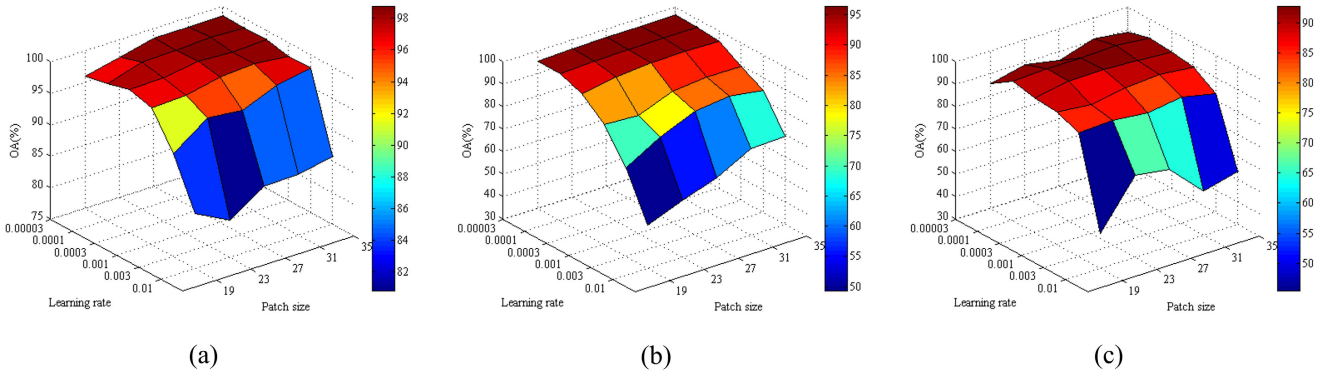


Fig. 9. OA (%) of GMA-Nets with different learning rates and patch sizes on the UP, SA, and HO images. (a) UP. (b) SA. (c) HO.

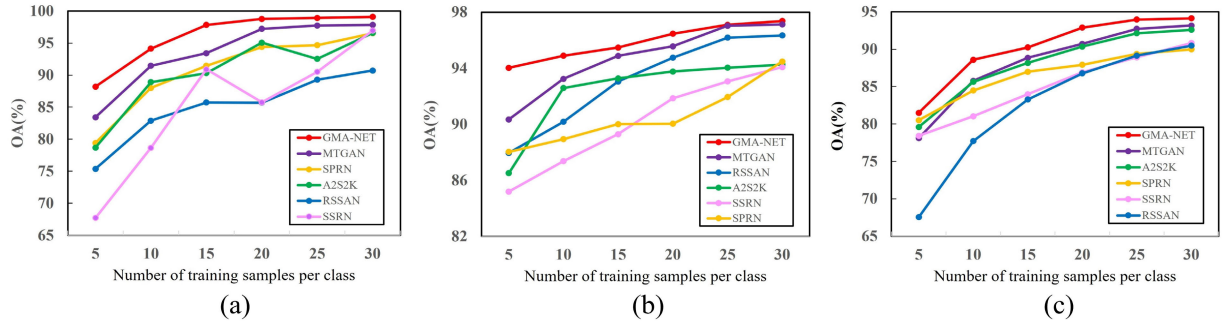


Fig. 10. OA (%) achieved by different methods on three HSIs when different numbers of training samples are given. (a) UP image. (b) SA image. (c) HO image.

TABLE IV
COMPUTATIONAL COMPLEXITY ANALYSIS OF DIFFERENT METHODS ON THE UP DATASET

Methods	Trainable Params(M)	Computation Cost(MFLOPs)	Training Time(s)	Testing Time(s)
SSRN	0.22	81.21	730	2.4
RSSAN	0.15	19.87	1377	2.6
A2S2K	0.22	131.68	1530	5.9
MTGAN	0.36	30.90	1478	3.2
SPRN	0.18	8.84	560	1.1
GMA-Net	1.29	181.97	1967	8.3

Intel Xeon(R) Gold 6148 CPU and an NVIDIA GeForce RTX 3090 GPU. Moreover, the training and testing samples are fixed for all the methods. The detailed values are demonstrated in Table IV.

It can be observed in Table IV that the proposed method achieves the largest values in terms of trainable parameters, MFLOPs, training time, and testing time, which indicates the higher complexity of the proposed method. It is because the proposed GMA-Net has a two-branch architecture which introduces more trainable parameters. Fortunately, in terms of training and testing time, the gap between the proposed method and other state-of-the-art methods is limited. For example, the training time of GMA-Net is 1967 s, while the training time of several methods (including A2S2K, MTGAN, and RSSAN) is about 1500 s. In other words, the only gap is less than 500 s, which is acceptable since training operations are off-line. The same phenomenon occurs in case of the testing time, e.g., the gaps are almost less than 6 s. Therefore, GMA-Net achieves the best classification performance with a not very high complexity, especially for running time.

IV. CONCLUSION

In this work, we propose an effective GMA-Net for HSI classification, which aims to jointly exploit the spectral reflectance characteristics of image and promising properties of the attention mechanism to enhance feature discrimination. This is achieved by designing a band-grouping strategy according to spectral correlation, followed by embedding spectral-spatial multi-attention module in CNN. Considering group-specific spectral-spatial features are sensitive to classify different materials, the relevance information of similar bands within each group are analyzed while the intergroup differential information is propagated in the proposed multi-attention module. By this way, the network can pay more attention to useful spectral and spatial information, leading to effectively learn highly discriminative spectral-spatial features in a band-group manner. Moreover, a two-branch architecture is also adopted in this network, where pixelwise spectral feature and patchwise spectral-spatial features are extracted and fused to further improve the classification results. Based on the proposed network, we train it using a small number

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of samples, and the experimental results on three well-known HSI datasets show that the proposed method can exhibit better classification performance than several state-of-the-art classification methods.

CNNs tend to extract local information of HSIs and are not good at modeling the long-term dependency of HSIs (i.e., the global information). Therefore, our future work will focus on coupling CNN with Transformer since the latter can effectively capture the long-term dependency and help improve the classification performance.

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